

CHAPTER 4

Activities & Lab Packet

Waves

A complete set of labs, worksheets, writing prompts, discussions, and group projects to accompany Chapter 4. Works with both the Standard and Easy Read versions of the textbook.



Georgia Standard S8P4

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1 Slinky Wave Lab

LAB

OBJECTIVE

Create and compare transverse and longitudinal waves.

TIME

25–30 minutes

STANDARDS

S8P4.a, S8P4.f

Materials (per group of 2)

- 1 Slinky (the long metal kind works best, but plastic works too)
- 1 piece of colored ribbon or yarn (about 2 inches)
- Open floor space (about 3 meters)
- Student data sheet

Setup

Tie the colored ribbon onto a coil in the MIDDLE of the Slinky. This will help students track how the Slinky's particles actually move during each type of wave. One student holds each end. Stretch it out on the floor — don't lift it up!

Student Instructions

1. **Make a transverse wave:** One partner holds their end still. The other partner quickly shakes their end SIDE-TO-SIDE (left-right) once. Watch what happens! Watch the ribbon especially.
2. Record: which direction does the wave TRAVEL? Which direction does the RIBBON move?
3. **Make a longitudinal wave:** Keep the Slinky stretched. Now PUSH your end toward your partner (a quick push and pull). Watch closely.
4. Record: which direction does the wave TRAVEL? Which direction does the RIBBON move?
5. **Test wavelength & frequency:** Shake your end side-to-side SLOWLY then FAST. What changes about the waves you make?
6. **Test amplitude:** Shake side-to-side with a SMALL motion, then a BIG motion. What changes?

TEACHER NOTES

What students should see:

- **Transverse:** Wave moves horizontally down the Slinky. Ribbon moves perpendicular (side to side). These are like light waves and water surface waves.
- **Longitudinal:** Wave moves horizontally. Ribbon also moves horizontally (parallel to the wave). Visible compressions and rarefactions. These are like sound waves.
- **Faster shake = higher frequency = shorter wavelength.** Slower shake = lower frequency = longer wavelength.
- **Bigger shake = bigger amplitude = more energy.**

Pro tip: Use a hard floor — carpet absorbs too much energy. Keep the Slinky stretched but not maxed out.

Safety: Don't lift the Slinky off the floor — they tangle terribly when airborne!

Slinky Wave Lab — Data Sheet

Name: _____

Period: _____

Part A: Compare the two types of waves

	Transverse	Longitudinal
How did you make it?		
Direction wave traveled		
Direction ribbon moved		
What it looked like		

Part B: Wave Properties

When I shook the Slinky FASTER, the wavelength got: ☐ longer ☐ shorter

When I shook the Slinky FASTER, the frequency got: ☐ higher ☐ lower

When I shook the Slinky with a BIGGER motion, the amplitude got: ☐ bigger ☐ smaller

Which type of Slinky wave is most like SOUND? Why?

Which type is most like LIGHT? Why?

2 String Phone Sound Lab

LAB

OBJECTIVE

Show that sound travels as vibrations through a medium.

TIME

30 minutes

STANDARDS

S8P4.a, S8P4.e

Materials (per pair)

- 2 paper cups, plastic cups, or empty tin cans
- 3–5 meters of string or kite line (test with different lengths)
- 1 pushpin or toothpick (to poke holes)
- 1 paperclip (to anchor the string ends inside the cups)
- Optional: a piece of yarn, fishing line, or rubber band to test as alternates

Setup

Poke a small hole in the bottom of each cup. Thread one end of the string through each cup from the outside in. Tie a paperclip to each end inside the cup to keep the string from pulling back through.

Student Instructions

1. One partner holds one cup up to their ear. The other partner holds their cup to their mouth.
2. Pull the string **TIGHT** between you. (This is the key step!)
3. Speak softly into your cup. Your partner should hear you clearly through their cup.
4. Now let the string go **SLACK** (loose). Try to talk through it. Notice the difference.
5. Try different sounds: whisper, sing, hum, laugh. Which travels best?
6. Try replacing the string with a piece of yarn, then a rubber band. Which works best?
7. **Big test:** Have a third person grab the string in the middle. Does the sound still come through? Why or why not?

TEACHER NOTES

The science: Your voice vibrates the air in the cup. The cup bottom vibrates. Those vibrations travel down the **TIGHT** string (the medium). At the other end, the cup bottom vibrates, then the air in the cup vibrates, then your friend's eardrum vibrates.

Why "tight" matters: Sound needs a medium with connected molecules. A slack string is loose — vibrations get absorbed. A tight string keeps the molecules close together, so the wave can travel.

The "grab the string" test: When someone grabs the string, they absorb the vibrations and the sound dies. This shows how a medium matters for mechanical waves.

Extension comparison: Discuss why this couldn't work in space — there's no medium for the sound to travel through between the cups.

String Phone Sound Lab — Observations

Name: _____

Period: _____

Tests & Results:

Test	How well could you hear? (1 = poorly, 5 = perfectly)
String tight, normal speaking	
String slack (loose)	
String tight, whisper	
Yarn instead of string	
Rubber band instead of string	
String tight + someone grabs it in the middle	

Why does the string have to be TIGHT for the phone to work?

What is the MEDIUM that sound travels through in this experiment?

Why does the sound STOP when someone grabs the middle of the string?

Could two astronauts use this phone in space? Why or why not?

3 Bending Light — Refraction Lab

LAB

OBJECTIVE

Demonstrate that light bends (refracts) when entering different mediums.

TIME

20–25 minutes

STANDARDS

S8P4.d, S8P4.g

Materials (per group)

- 1 clear glass cup or jar (smooth-sided)
- Water (enough to fill the cup almost full)
- 1 pencil or straw
- 1 coin (penny works great)
- 1 piece of white paper
- 1 sheet of paper with a thick arrow drawn on it (like 1 inch long)

Three Quick Experiments

Experiment A — The Broken Pencil

1. Fill the cup with water about $\frac{3}{4}$ full.
2. Put a pencil straight down into the water so half is in, half is out.
3. Look from the side. The pencil should look **BROKEN** or **BENT** at the water's surface!
4. Take it out. Is the pencil actually broken? (No!)

Experiment B — The Magic Coin

1. Empty the cup. Place a coin on the bottom of the empty cup.
2. Back up until the coin is **JUST** barely hidden behind the rim of the cup (you can't see it).
3. **WITHOUT** moving your head, ask your partner to slowly pour water into the cup.
4. The coin should appear to "rise" into view! Magic? No — refraction!

Experiment C — The Reversed Arrow

1. Empty and dry your cup. Stand the paper with the arrow about 6 inches behind the cup.
2. Look at the arrow through the cup. It looks normal.
3. Now fill the cup with water and look again. The arrow should appear to **REVERSE** direction!

TEACHER NOTES

What's happening:

- **Broken pencil:** Light bends when going from water to air, so the part of the pencil in the water appears in a different position.
- **Magic coin:** Water bends the light coming from the coin, so it appears to be in a different spot than it actually is.
- **Reversed arrow:** The water-filled cup acts like a **CONVEX LENS**! Light passes through and crosses over, flipping the image — same principle as your eye's lens, magnifying glasses, and cameras.

Tie back to the textbook: A convex lens (fatter in the middle, like the cup of water) bends light to form an image. This is exactly how Section 4.10 talks about lenses!

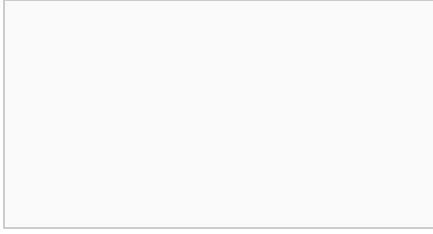
Refraction Lab — Observations

Name: _____

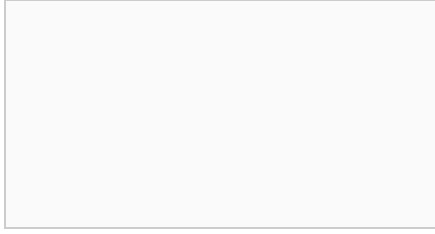
Period: _____

Draw what you observed in each experiment:

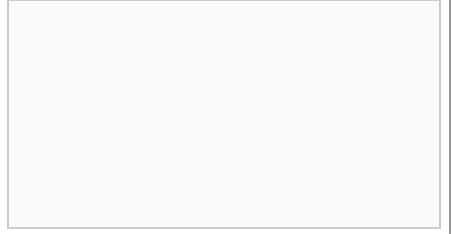
A: Broken Pencil



B: Magic Coin



C: Reversed Arrow



What is REFRACTION? Write a one-sentence definition.

Why did the arrow appear to flip when you looked through the water?

A magnifying glass uses a CONVEX lens. Why is the cup of water sort of acting like one?

4 Wave Anatomy Practice

WORKSHEET

OBJECTIVE

Identify parts of a wave and compare wave properties.

TIME

15–20 minutes

STANDARDS

S8P4.f

Wave Anatomy

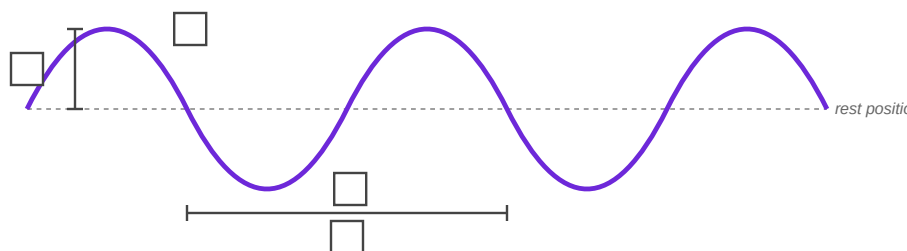
Label the parts and compare wave properties.

Name: _____

Period: _____

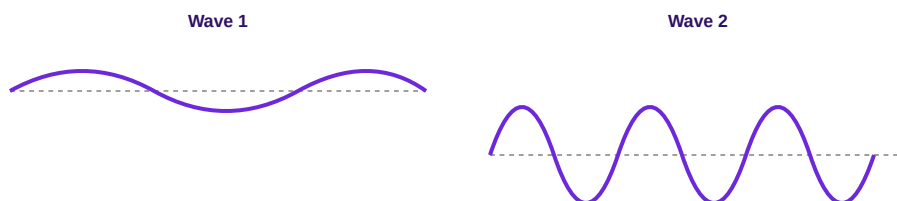
Part A: Label this wave

Write the letters A, B, C, D in the right spots: A = Crest B = Trough C = Wavelength D = Amplitude



Part B: Compare the waves

Below are two waves drawn at the same scale.



1. Which wave has a HIGHER frequency? → **Wave 1 / Wave 2**
2. Which wave has a LONGER wavelength? → **Wave 1 / Wave 2**
3. Which wave has a BIGGER amplitude? → **Wave 1 / Wave 2**
4. Which wave has MORE energy? → **Wave 1 / Wave 2**

Part C: Apply It

5. If a wave's amplitude gets bigger, its energy goes: **UP / DOWN / SAME**
6. If frequency goes UP, wavelength goes: **UP / DOWN / SAME**
7. A loud song has a ____ amplitude than a quiet song. **BIGGER / SMALLER**
8. A high-pitched sound has a ____ frequency than a low-pitched sound. **HIGHER / LOWER**

ANSWER KEY

Part A: A = top of wave ($y \approx 40$), B = bottom of wave ($y \approx 140$), C = between adjacent crests, D = from rest line up to crest

Part B: 1) Wave 2 (more cycles in same space) | 2) Wave 1 (longer between crests) | 3) Wave 1 (taller) | 4) Tricky! Both have energy from their properties. Accept either with reasoning — Wave 2 has higher frequency, Wave 1 has higher amplitude.

Part C: 5) UP | 6) DOWN | 7) BIGGER | 8) HIGHER

5 EM Spectrum & Energy

WORKSHEET

OBJECTIVE

Order the EM spectrum and connect each type to real-world uses.

TIME

15–20 minutes

STANDARDS

S8P4.b, S8P4.c

EM Spectrum & Energy

The EM spectrum from LOW energy to HIGH energy.

Name: _____

Period: _____

Part A: Put these in the right order from LOWEST to HIGHEST energy.

Word bank: Visible Light, X-Rays, Radio Waves, Gamma Rays, Microwaves, Ultraviolet (UV), Infrared

1. (lowest energy): _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. (highest energy): _____

Part B: Match each type to its real-world use.

EM Wave Type	Used For...
1. Radio waves	A. Heating leftovers, cell phone signals
2. Microwaves	B. The only kind your eyes can see
3. Infrared	C. Cancer treatment, made by nuclear reactions
4. Visible light	D. TV broadcasts, AM/FM radio
5. Ultraviolet (UV)	E. TV remotes, sensing body heat, night vision
6. X-rays	F. Sunburns, killing bacteria
7. Gamma rays	G. Seeing bones inside the body

Write your answers: 1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____ 7. _____

Part C: True or False

1. All EM waves travel at the same speed in empty space. _____

2. EM waves need a medium (like air) to travel. _____

3. Radio waves have more energy than X-rays. _____

4. Visible light is only a TINY part of the EM spectrum. _____

ANSWER KEY

Part A (low to high): 1. Radio | 2. Microwave | 3. Infrared | 4. Visible Light | 5. UV | 6. X-rays | 7. Gamma rays

Part B: 1-D | 2-A | 3-E | 4-B | 5-F | 6-G | 7-C

Part C: 1) T | 2) F (no medium needed!) | 3) F (X-rays have MUCH more energy) | 4) T

6 Wave Behaviors Sort

WORKSHEET

OBJECTIVE

Identify reflection, refraction, absorption, diffraction, and transmission in real life.

TIME

15 minutes

STANDARDS

S8P4.d

Wave Behaviors Sort

Decide which wave behavior is happening. Write your answer in the blank.

Name: _____

Period: _____

REFLECTION	REFRACTION	ABSORPTION	DIFFRACTION	TRANSMISSION
bounces off	bends	soaked up	bends around	passes through

#	Situation	Behavior
1	You look at yourself in a mirror.	
2	A straw in a glass of water looks bent at the surface.	
3	A black T-shirt feels hot in the sunlight.	
4	Sunlight passes through a clean glass window.	
5	You hear your friends talking around the corner before you see them.	
6	An echo bounces back from a cliff wall.	
7	A magnifying glass makes letters look bigger.	
8	You shout into a canyon and hear yourself a moment later.	
9	Light passes through clear water to show fish underneath.	
10	Sound waves bend around a doorframe.	
11	Sunlight is "soaked up" by a solar panel.	
12	Light bends as it enters a swimming pool from the air.	

Short Answer: Why can you hear someone talking around a corner, but you can't see them around the corner?

ANSWER KEY

1) Reflection | 2) Refraction | 3) Absorption | 4) Transmission | 5) Diffraction | 6) Reflection | 7) Refraction (lens bends light) | 8) Reflection | 9) Transmission | 10) Diffraction | 11) Absorption | 12) Refraction

Short answer: Sound waves have a long wavelength and can diffract (bend) around objects like doorframes. Light has a very short wavelength and doesn't diffract much around large objects, so we can't see around corners.

7 Writing Prompts

WRITING

OBJECTIVE

Use scientific vocabulary to explain wave concepts in writing.

TIME

15–30 minutes each

STANDARDS

Cross-cutting

Level 1 — Beginner (1 paragraph)

1. **Wave in the Stadium.** The textbook compares waves to a stadium wave. In your own words, explain what this means. Why doesn't the matter (or people) actually travel with the wave?
2. **Pitch & Volume.** Pick your favorite song. Describe parts where the pitch is high vs. low and parts where the volume is loud vs. quiet. Use the words "frequency" and "amplitude."
3. **Why Space Is Silent.** In sci-fi movies, you hear explosions in space. Explain why this is scientifically wrong using what you know about mechanical waves.

Level 2 — Intermediate (3 paragraphs)

4. **The Sound Journey.** Trace the path of a sound wave from a guitar string to your ear. Include vibration, air molecules, the medium, your eardrum, and what your brain does at the end.
5. **EM Spectrum Tour.** You're a tour guide leading visitors through the EM spectrum. Describe the seven main "stops" in order (low to high energy), and one important use at each one.
6. **Why Sunglasses Work.** Explain how sunglasses use absorption to protect your eyes. Why are darker lenses more effective? Why are UV-blocking lenses important even on cloudy days?

Level 3 — Advanced (essay)

7. **The Phone in Your Pocket.** Your smartphone uses at least four different types of EM waves. Identify them and explain what each is used for. Then argue: which one would be the LEAST replaceable if it suddenly stopped working?
8. **Mechanical vs. EM Debate.** Some scientists argue that EM waves are the "more useful" kind of wave because they don't need a medium. Others argue mechanical waves are more important because they let us hear and feel. Pick a side and defend it with at least three examples.

8 Classroom Discussion Questions

DISCUSSION

OBJECTIVE

Practice scientific thinking through verbal reasoning.

USE

Warm-ups, exit tickets, Socratic seminar

STANDARDS

Cross-cutting

TEACHER NOTES

Use think-pair-share format. Many of these questions are great for sparking debate because the answers are surprising. Push students to use evidence from the chapter, not just guesses.

Quick Warm-Ups (1–2 min each)

1. Do waves carry matter or energy? How do you know?
2. Could a fish in a fishbowl hear you? Why or why not?
3. Is the wave at a stadium more like a sound wave or a light wave?
4. Can a wave bend? Give an example.
5. What color of visible light has the most energy?

Deeper Discussions (5–15 min each)

6. **The Silent Movie.** If you were on the Moon and someone exploded a firework, would you see it? Hear it? Why or why not? Connect to the difference between mechanical and EM waves.
7. **The Disco Ball Problem.** A disco ball has hundreds of tiny mirrors. When light hits it, you see spots of light everywhere. Use the concept of reflection to explain why this happens.
8. **Why the Sky Is Blue.** Discuss: as sunlight passes through Earth's atmosphere, blue light scatters (a type of refraction/diffraction) more than other colors. Why does this make the sky look blue? Why are sunsets red?
9. **The Whale Question.** Whales communicate over hundreds of miles in the ocean. Sound travels MUCH faster underwater than in air. Discuss why this is — and what this tells us about how density affects waves.
10. **X-Ray Vision?** X-rays pass through your skin but not your bones, which is why we use them for medical imaging. Discuss: why does this work? What does it tell us about absorption?

9 Design a Musical Instrument

GROUP PROJECT

OBJECTIVE

Design and build an instrument that demonstrates pitch (frequency) and volume (amplitude).

TIME

2–3 class periods

STANDARDS

S8P4.f, S8P4.a

Project Description

Each group designs and builds their own musical instrument using everyday materials. The instrument must produce **at least 3 different pitches** (frequencies) and demonstrate the ability to play **at different volumes** (amplitudes).

Requirements

- The instrument must produce at least **3 distinct notes**
- Must be able to play softer AND louder
- Made from recycled or common household materials
- Group must explain HOW their instrument creates different frequencies
- Group must perform a simple melody (3+ notes) for the class

Instrument Type Ideas

Instrument Type	Materials & How It Works
String Instrument	Rubber bands on a tissue box. Tighter bands = higher pitch. Pluck harder = louder.
Wind Instrument	Glass bottles with different water levels. Less water = lower pitch when blowing across the top.
Percussion	Glass cups filled with different amounts of water. Tap with a spoon. Different levels = different pitches.
Pan flute	Straws cut to different lengths and taped together. Shorter straws = higher pitch.
Box guitar	Shoebox with rubber bands of different thicknesses stretched across an opening.

Required Deliverables

1. **Build the instrument.**
2. **Label the parts** — show which part vibrates, which part amplifies, and how pitch changes.
3. **Written explanation:** How does your instrument change the frequency to make different pitches? How does it change amplitude to make different volumes?
4. **Performance:** Play a simple melody (Mary Had a Little Lamb, Twinkle Twinkle, etc.) or your own original 3-note song for the class.

TEACHER NOTES

The key learning: Whatever the instrument does to change pitch, the physics is the same — it's changing the FREQUENCY of the sound wave. Tighter string, shorter air column, smaller glass — they all vibrate faster, creating higher frequency.

Differentiation: For struggling readers, allow them to choose from the suggested types. For advanced students, encourage them to make a more original design.

Rubric (suggested): Instrument works (30%), produces 3+ pitches (20%), explanation accuracy (25%), performance (15%), creativity (10%).

Safety: If using glass bottles, set ground rules about handling carefully and not tapping too hard.

10 Wave Vocab Pictionary

REVIEW GAME

OBJECTIVE

Review chapter vocabulary in an active, visual way.

TIME

20–30 minutes

STANDARDS

S8P4 cross-cutting

How to Play

Print and cut out the cards below. Divide the class into two teams. One person draws a card and has 60 seconds to draw the term on the board **WITHOUT** using words, letters, or numbers. Their team has to guess. If they guess correctly, they earn a point. Most points at the end wins.

Variations

- **Charades mode:** Act it out instead of drawing.
- **Telephone Pictionary:** First person draws, second person describes the drawing in words on a new paper, third person draws THAT description — pass through the team!
- **Draw & Define:** After someone guesses correctly, the drawer must give the definition of the word to earn the point.

Pictionary Card Set — Print & Cut

WAVE	CREST	TROUGH
WAVELENGTH	AMPLITUDE	FREQUENCY
TRANSVERSE WAVE	LONGITUDINAL WAVE	COMPRESSION
MEDIUM	MECHANICAL WAVE	EM WAVE
REFLECTION	REFRACTION	ABSORPTION
DIFFRACTION	TRANSMISSION	PITCH
VOLUME	ECHO	VISIBLE LIGHT
RADIO WAVES	X-RAYS	CONVEX LENS

TEACHER NOTES

Easier-to-draw cards: CREST, TROUGH, WAVELENGTH, AMPLITUDE, MEDIUM, REFLECTION, REFRACTION, CONVEX LENS — these have visual representations students saw in the textbook diagrams.

Harder-to-draw cards: FREQUENCY, EM WAVE, PITCH, VOLUME — encourage stick figures, arrows, and symbols.

Visual scaffolding: Keep the textbook open during the game so students can reference the figures. For struggling readers, allow them to peek at the chapter while drawing.

Quick prizes: Homework pass, free seat choice, or candy.